Danmarks Tekniske Universitet

DTU

# 42114 - Integer Programming Assignment 1


**Authors**

Haukur Methúsalem Óskarsson - s253131
Hjörtur Bruun - s263451
Hrannar Ingi Arnarsson - s252867
Kolbeinn Tumi Sindrason - s253026


November 4, 2026

## Modelling: Question 1

We assume that each surgery is performed in full on a single day, that $r_s \in D$, and that $p_s, t_s, H_d \geq 0$. Let $x_{sd} \in \{0,1\}$ equal 1 if surgery $s$ is performed on day $d$, and let $L_s \geq 0$ be the number of days surgery $s$ is performed after $r_s$.

$$\min \quad \sum_{s \in S} p_s L_s \tag{1}$$

$$\text{s.t.} \quad \sum_{d \in D} x_{sd} = 1 \qquad \forall s \in S \tag{2}$$

$$\sum_{s \in S} t_s x_{sd} \leq H_d \qquad \forall d \in D \tag{3}$$

$$L_s \geq (d - r_s) - M(1 - x_{sd}) \qquad \forall s \in S,\ d \in D \tag{4}$$

$$x_{sd} \in \{0,1\},\ L_s \geq 0 \qquad \forall s \in S,\ d \in D \tag{5}$$

(2) schedules every surgery exactly once and (3) enforces the daily capacity. In (4), $x_{sd} = 1$ gives $L_s \geq d - r_s$, while $x_{sd} = 0$ makes the right-hand side non-positive so the constraint is inactive. Minimisation then gives $L_s = \max\{0, d_s - r_s\}$, where $d_s$ is the day assigned to $s$. Since $d - r_s \leq n - 1$, the value $M = n - 1$ is valid.

## Question 2

Add $w_{kd} \in \{0,1\}$, equal to 1 if surgeon $k$ works on day $d$, and the constraints

$$x_{sd} \leq w_{kd} \qquad \forall k \in K,\ s \in S_k,\ d \in D \tag{6}$$

$$w_{kd} + w_{k,d+1} + w_{k,d+2} \leq 2 \qquad \forall k \in K,\ d = 1, \ldots, n-2 \tag{7}$$

(6) marks a surgeon as working on every day they operate, and (7) forbids three consecutive working days. Equivalently, (6) can be written in Big-M form as $\sum_{s \in S_k} x_{sd} \leq M_k\, w_{kd}$ for all $k \in K$, $d \in D$, $M_k = |S_k|$, since surgeon $k$ can perform at most $|S_k|$ surgeries on a single day.

## Question 3

Add $z_d \in \{0,1\}$, equal to 1 if at least $\rho$ hours are kept free on day $d$, replace (3) by (8), and add (9):

$$\sum_{s \in S} t_s x_{sd} \leq H_d - \rho + M(1 - z_d) \qquad \forall d \in D \tag{8}$$

$$\sum_{d \in D} z_d \geq \lceil 0.8\, n \rceil \tag{9}$$

If $z_d = 1$, at least $\rho$ hours are unused on day $d$. If $z_d = 0$, the constraint must reduce to the ordinary capacity $H_d$, which requires $M = \rho$. With this choice (8) becomes $\sum_s t_s x_{sd} \leq H_d - \rho z_d$. Constraint (9) ensures that capacity is reserved on at least 80% of the days.

**Question 4**

Add $y_d \in \{0,1\}$, equal to 1 if the robot is activated on day $d$, replace (1) by (10), and add (11):

$$\min \quad \sum_{s\in S} p_s L_s + \gamma \sum_{d\in D} y_d \tag{10}$$

$$x_{sd} \le y_d \qquad \forall s \in R,\ d \in D \tag{11}$$

(11) activates the robot on every day with a robotic surgery, and since $\gamma > 0$, $y_d = 0$ on all other days. Equivalently, in Big-M form: $\sum_{s\in R} x_{sd} \le |R|\, y_d$ for all $d \in D$.

**Lagrangian Relaxation: Question 5**

Let $I = \{1,\dots,15\}$ be the set of initiatives and $J = \{1,2,3\}$ the set of teams. For initiative $i \in I$, $b_i$ is its strategic benefit and $a_i$ its resource consumption (Table 1). Team $j \in J$ has capacity $C_j$, with $C_1 = 28$, $C_2 = 32$ and $C_3 = 35$. Let

$$P = \{(2,1),(3,1),(4,1),(7,5),(8,5),(9,6),(10,6),(11,7),(12,8),(15,10)\}$$

be the set of precedence pairs, where $(i,k) \in P$ means that initiative $i$ requires initiative $k$.

Let $x_{ij} \in \{0,1\}$ equal 1 if initiative $i$ is selected and assigned to team $j$, and $y_i \in \{0,1\}$ equal 1 if initiative $i$ is selected. So the model is:

$$\max \quad \sum_{i\in I} b_i\, y_i \tag{12}$$

$$\text{s.t.} \quad \sum_{j\in J} x_{ij} = y_i \qquad \forall i \in I \tag{13}$$

$$\sum_{i\in I} a_i\, x_{ij} \le C_j \qquad \forall j \in J \tag{14}$$

$$y_i \le y_k \qquad \forall (i,k) \in P \tag{15}$$

$$y_{11} + y_{12} \le 1 \tag{16}$$

$$y_2 + y_3 + y_4 + y_7 + y_8 = 3 \tag{17}$$

$$x_{ij},\ y_i \in \{0,1\} \qquad \forall i \in I,\ j \in J \tag{18}$$

(12) maximises the total strategic benefit. (13) assigns every selected initiative to exactly one team and unselected initiatives to none. (14) enforces the team capacities. (15) ensures that an initiative is only selected if its prerequisite is selected. (16) is the exclusivity requirement, and (17) selects exactly three initiatives from $\{2,3,4,7,8\}$.

Without the side constraints (15)–(17), the model is exactly the *Multiple Knapsack Problem* (MKP): each initiative is an item with profit $b_i$ and weight $a_i$, each team is a knapsack with capacity $C_j$, and each item is placed in at most one knapsack. The total resource consumption of all initiatives is 148, while the total capacity is only $28+32+35 = 95$, so the capacity constraints are binding and a selection has to be made. Unlike the Generalized

Assignment Problem, items need not be assigned, and profits and weights do not depend on the team. The problem is therefore a Multiple Knapsack Problem with additional precedence, conflict and cardinality side constraints.

**Question 6**

We relax the assignment constraints (13), which are the only constraints linking the selection variables $y$ to the team variables $x$, and keep the capacity constraints (14) and the side constraints (15)–(17). Since (13) are equalities, the multipliers $u_i \in \mathbb{R}$, $i \in I$, are free. The relaxed model is

$$
\begin{aligned}
Z_{LR}(u) = \max \quad & \sum_{i\in I} b_i y_i + \sum_{i\in I} u_i \Big( \sum_{j\in J} x_{ij} - y_i \Big) && && (19)\\
\text{s.t.} \quad & \sum_{i\in I} a_i x_{ij} \le C_j && \forall j \in J \\
& y_i \le y_k && \forall (i,k) \in P \\
& y_{11} + y_{12} \le 1, \qquad y_2 + y_3 + y_4 + y_7 + y_8 = 3 \\
& x_{ij},\ y_i \in \{0,1\} && \forall i \in I,\ j \in J
\end{aligned}
$$

Every feasible solution of the original problem satisfies $\sum_j x_{ij} - y_i = 0$, so the penalty term vanishes and $Z_{LR}(u) \ge Z_{IP}$ for all $u \in \mathbb{R}^{15}$.

Collecting terms, no constraint contains both $x$ and $y$, and the relaxation decomposes into independent subproblems:

$$
Z_{LR}(u) = \underbrace{\max_{y\in Y} \sum_{i\in I} (b_i - u_i)\, y_i}_{\text{selection problem}} + \sum_{j\in J} \underbrace{\max \Big\{ \sum_{i\in I} u_i x_{ij} \ :\ \sum_{i\in I} a_i x_{ij} \le C_j,\ x_{ij} \in \{0,1\} \Big\}}_{\text{0-1 knapsack for team } j},
$$

where $Y$ is the set of binary $y$ satisfying (15)–(17). The multiplier $u_i$ can be read as a price on initiative $i$: the selection problem pays $u_i$ for selecting it, and each team is paid $u_i$ for packing it. The three knapsacks are solved by dynamic programming, and the selection problem has only 15 binary variables.

The relaxation is chosen so that the subproblem is easy but does *not* have the integrality property: the LP relaxation of a knapsack can have fractional solutions, and the side constraints are not integral either. The Lagrangian dual bound is $Z_{LD} = \max\{b^\top y : (13),\ (x,y) \in \mathrm{conv}(X_1 \times X_2 \times X_3 \times Y)\}$, where $X_j$ is the set of feasible packings of team $j$, so it can be strictly tighter than the LP bound, $Z_{IP} \le Z_{LD} \le Z_{LP}$. Relaxing instead the side constraints keeps the hard Multiple Knapsack structure in the subproblem and convexifies only the capacity constraints. We tested this alternative, and it gives only $Z_{LD} = 254.73$.

### Question 7

The Lagrangian dual problem finds the multipliers that give the tightest (smallest) upper bound from the relaxation in Question 6:

$$Z_{LD} = \min_{u \in \mathbb{R}^{15}} Z_{LR}(u) = \min_{u \in \mathbb{R}^{15}} \left\{ \max_{y \in Y} \sum_{i \in I} (b_i - u_i)\, y_i + \sum_{j \in J} \max_{x_{\cdot j} \in X_j} \sum_{i \in I} u_i x_{ij} \right\}. \tag{20}$$

The multipliers are unrestricted in sign because the relaxed constraints are equalities.

$Z_{LR}$ is the maximum of a finite number of functions that are linear in $u$, one for each feasible $(x, y)$. It is therefore convex and piecewise linear, but not differentiable everywhere, which is why we use a subgradient method to solve (20). For a subproblem solution $(x^k, y^k)$ at $u^k$, the vector $g^k$ with $g_i^k = \sum_j x_{ij}^k - y_i^k$ is a subgradient of $Z_{LR}$ at $u^k$.

### Question 8

The LP relaxation of the model in Question 5 and the Lagrangian dual from Question 7 were solved in Julia/JuMP with HiGHS (`assignment_relaxation.jl`). In each iteration the three knapsacks are solved by dynamic programming and the selection problem as a small MIP. The multipliers start at $u^0 = 0$ and are updated as

$$u^{k+1} = u^k - t_k\, g^k, \qquad t_k = \theta_k \frac{Z_{LR}(u^k) - \underline{Z}}{\|g^k\|^2},$$

without projection, since $u$ is free. If initiative $i$ is selected but not packed ($g_i < 0$), its price $u_i$ increases, which makes packing it more attractive for the teams and selecting it less attractive. If it is packed more often than selected ($g_i > 0$), the price decreases. $\underline{Z}$ is the best feasible value found so far by the Lagrangian heuristic of Question 9, and $\theta_0 = 2$ is halved after 10 iterations without improvement. Since all benefits are integer, the method stops when $\lfloor \min_k Z_{LR}(u^k) \rfloor \le \underline{Z}$.

| Bound | Value |
|---|---|
| Optimal IP value $Z_{IP}$ | 244.00 |
| Lagrangian bound, subgradient (58 iterations) | 244.90 |
| Lagrangian dual $Z_{LD}$ (exact, cutting planes) | 244.00 |
| LP relaxation bound $Z_{LP}$ | 254.89 |

The best upper bound from the subgradient method is 244.90, which gives the integer bound 244. The exact value of the Lagrangian dual, computed with a cutting-plane method in `LB.jl`, is $Z_{LD} = 244$. The Lagrangian bound is therefore much tighter than the LP bound of 254.89 and closes the entire gap to $Z_{IP}$.

This agrees with $Z_{IP} \le Z_{LD} \le Z_{LP}$. The LP relaxation only uses the LP versions of the capacity and side constraints, whereas the Lagrangian dual uses the convex hulls of the knapsack sets $X_j$ and of the selection set $Y$. Since none of these has the integrality

property, the convex hulls are strictly smaller than their LP relaxations, and in this instance enough smaller that the bound equals the optimum. For comparison, relaxing the side constraints gives only 254.73. Relaxing the side constraints together with the assignment or capacity constraints gives exactly $Z_{LP}$, since the remaining subproblem then has the integrality property.

**Question 9**

The Lagrangian relaxation does *not* terminate with a feasible solution. In the final iteration ($k = 58$, $Z_{LR} = 244.90$) the selection problem chooses

$$y = \{1, 2, 3, 4, 6, 9, 10, 13, 14, 15\}, \qquad \text{benefit } 244,$$

which satisfies all side constraints, but the knapsacks pack the items in Table 1. Initiative 6 is packed by all three teams, 14 by two teams, and 2, 3, 4 and 10 are selected but not packed, so the assignment constraints are violated.

| Team (capacity) | Initiatives packed | Load |
|---|---|---|
| 1 (28) | 6, 9, 14 | 28 |
| 2 (32) | 6, 13, 15 | 32 |
| 3 (35) | 1, 6, 14 | 35 |

Table 1: Knapsack solutions in the final subgradient iteration.

**Lagrangian heuristic.** The selection $y$ already satisfies the side constraints, so only the packing has to be repaired:

1. **Keep:** for every selected initiative packed by at least one team, keep it in one of those teams. Remove duplicates and initiatives that are packed but not selected.
2. **Place:** place the remaining selected initiatives, largest $a_i$ first, in the team with the most free capacity. If one does not fit, drop the *removable* initiative with the lowest $b_i/a_i$ and try again. An initiative is removable if it is not in $\{2, 3, 4, 7, 8\}$ and not a prerequisite of another selected initiative, so dropping it keeps the precedence, exclusivity and cardinality constraints satisfied.
3. **Greedy fill:** add unselected initiatives outside $\{2, 3, 4, 7, 8\}$ in decreasing order of $b_i/a_i$ whenever they fit in a team and their prerequisites and exclusivity allow it.

Steps 1–3 never exceed a capacity and never violate a side constraint, so the result is feasible.

**Demonstration.**

1. The duplicates of 6 and 14 are removed. Team 1 keeps $\{6, 9, 14\}$ (load 28), team 2 keeps $\{13, 15\}$ (load 20) and team 3 keeps $\{1\}$ (load 14).

2. Initiatives 4, 2, 10 and 3 must be placed. Initiative 4 goes to team 3, 2 to team 2 and 10 to team 3, giving loads 28, 28 and 32. Initiative 3 ($a_3 = 7$) does not fit. The removable initiatives are 9, 13, 14 and 15, and 9 has the lowest ratio (22/7), so it is dropped from team 1 and initiative 3 is placed there instead.

3. The free capacities are 0, 4 and 3, so nothing can be added.

| Team (capacity) | Initiatives | Load |
|---|---|---|
| 1 (28) | 3, 6, 14 | 28 |
| 2 (32) | 2, 13, 15 | 28 |
| 3 (35) | 1, 4, 10 | 32 |

Table 2: Feasible solution from the Lagrangian heuristic in the final iteration (benefit 222).

The resulting solution in Table 2 is feasible with benefit 222. The heuristic is myopic: the selection $y$ in the final iteration is in fact optimal, and all ten initiatives fit exactly into the 95 units of capacity, but the largest-first placement does not find this packing. Applying the heuristic in every subgradient iteration found the values 199, 213, 222 and, in iteration 50, 244 with teams $\{2, 13, 15\}$, $\{1, 4, 10\}$ and $\{3, 6, 9, 14\}$ (loads 28, 32 and 35). Together with the Lagrangian bound,

$$244 \leq Z_{IP} \leq \lfloor 244.90 \rfloor = 244,$$

so the Lagrangian relaxation and the heuristic together prove that this solution is optimal.

# Question 10

**Greedy principle.** Starting from an empty selection, we repeatedly add the route with the lowest *effective cost per newly covered location*,

$$r_j = \frac{c_j + \sum_{i \in S_j \cap U} \rho_i}{|S_j \setminus U|},$$

where $U$ is the set of locations already covered. The numerator is the route cost plus the over-coverage penalties the route would add, and the denominator is the number of uncovered locations it covers. Only routes covering at least one new location are considered, and ties are broken by the lowest route index. The procedure stops when all ten locations are covered, so the solution is feasible.

**Iterations.**

| It. | Selected route | $r_j$ | Covered after iteration |
|---|---|---|---|
| 1 | S2 $\{1,3,4\}$ | $8/3 = 2.67$ | $\{1,3,4\}$ |
| 2 | S3 $\{2,6\}$ | $7/2 = 3.50$ | $\{1,2,3,4,6\}$ |
| 3 | S8 $\{8,9,10\}$ | $14/3 = 4.67$ | $\{1,2,3,4,6,8,9,10\}$ |
| 4 | S4 $\{3,7\}$ | $(6+1)/1 = 7$ | $\{1,\ldots,4,6,\ldots,10\}$ |
| 5 | S1 $\{1,2,5\}$ | $(9+2+1)/1 = 12$ | $\{1,\ldots,10\}$ |

**Solution.** The greedy solution is $\{S1, S2, S3, S4, S8\}$. Locations 1, 2 and 3 are each covered twice, and no route can be removed without leaving a location uncovered. The objective value is

$$\underbrace{9+8+7+6+14}_{\text{route costs}=44} + \underbrace{\rho_1+\rho_2+\rho_3}_{2+1+1=4} = 48.$$